

Compiling a Vitess Distribution

Vitess is distributed as an installation image for Windows (7 upwards), or tarballs for Linux / MacOS.

The installation files comprise sources, help files, documentation, and binaries in both cases. The Windows installation comprises binaries for Tcl/Tk and gnuplot, too.

To develop Vitess all files except binaries are kept in a **subversion repository**
<https://www.helmholtz-berlin.de/svn/vitess>

We assume to have checked out a working copy of the trunk, in the example at a Linux PC in the directory ~/SRC/vitess like

```
cd ~/SRC
```

```
svn co https://www.helmholtz-berlin.de/svn/vitess vitess
```

The following text describes what had been done to produce the distribution files for Vitess version 3.4.

Steps done for Vitess 3.4

First bring the working copy contents to an up to date state, then do on Linux

```
cd ~/SRC/vitess
```

```
svn update
```

```
svn commit -m 'Version 3.4 release candidate 3'
```

To prepare a distribution we create separate directories under Linux, to be filled with binaries for the different system Linux, MacOS, and Windows.

In the example these are

```
~/VitessModulesUnix
```

```
~/VitessModulesWindows
```

The compilation resources SRC/Makefile and SRC/vitess.mak may be prepared with the Perl script SRC/mmake.pl which should be augmented and adopted to reflect the dependencies for C sources.

Compiling Linux binaries

```
cd ~/SRC/vitess/trunk
```

Delete symbolic links in MODULES/ sub directory.

```
TOOLS/unsetLinks.pl ./
```

```
cd ~/SRC/vitess/trunk/SRC
```

Prepare the Makefile.

```
./mmake.pl
```

```
cd ~/SRC/vitess/trunk
make clean
make
```

The install step moves binaries to the MODULES/ sub directory .

```
make install
```

Move binaries from MODULES/ to ~/VitessModulesUnix .

```
cd ~/SRC/vitess/trunk
TOOLS/moveUnixBinaries.pl ./
```

Create symbolic links in MODULES/ to allow working with these.

```
TOOLS/setLinks.pl ./
```

Compiling macOS / Darwin binaries

We used a mini Mac called daisy, with a working copy ~/SRC/vitess .

First, if you have not done before, check out a working copy

```
cd ~/SRC
svn co http://www.helmholtz-berlin.de/svn/vitess/trunk vitess
```

Later on get the latest version:

```
cd ~/SRC/vitess
```

If daisy has a new svn version:

```
svn upgrade
```

Subversion needs a user, the password will be prompted;

The example user name for the repository is xyz

```
svn --username xyz update
cd SRC
```

```
./mmake.pl
```

```
make clean
```

```
make
```

```
make install
```

At last copy the Darwin binaries. We did it at the Linux PC, where xyz_local was the used account on daisy.

```
scp 'xyz_local@daisy:SRC/vitess/MODULES/*Darwin*' ~/VitessModulesUnix/
```

Compiling Windows binaries

To compile the Windows binaries we used a Windows 7 PC, which in the example has a working copy at C:\Benutzer\xyz\vitess , fetched from the repository with TOTOISESVN, and updated then.

Using a Visual Studio 2017 command window

```
cd C:\Benutzer\xyz\viteSS\SRC
nmake /f viteSS.mak all
```

The resulting .exe images are in the `release` sub directory, to be copied to
`~/ViteSSModulesWindows` on the Linux PC.

(Some old helper binaries are ShellExec.exe, gzip.exe and helpa.exe are not compiled anew. They are heritage.)

Prepare install files

In the example we use a clean export of the repository to a Unix home sub directory.

(Have a clean copy, not a working one with additional files and svn backup sub directories!)

```
cd ~/tmp/
```

Remove old contents if necessary:

```
rm -rf viteSS3.4/
```

Then export the clean copy

```
svn export http://www.helmholtz-berlin.de/svn/viteSS/trunk viteSS3.4
```

Compiling a self installing image with Inno Setup

In the example the virtual Windows 7 PC has access to the Unix home directory as network drive
`H:` to access all files to be compressed.

The windows binaries are copied with Linux

```
cp ~/ViteSSModulesWindows/*.exe ~/tmp/viteSS3.4/MODULES/
```

The Inno Setup definition file `TOOLS/Win7ViteSS3.4.iss` should be copied to an arbitrary directory on the Windows PC, and edited to reflect the local installation of Tcl/Tk and gnuplot.

The .iss file's resources, which should be present on the Windows PC:

`H:\tmp\viteSS3.4\*` the clean export copy of ViteSS

`C:\Tcl32\*` the directory where Tcl/Tk 8.5 has been installed

`C:\Program Files\gnuplot\*` a gnuplot installation directory

The `setup.exe` file is the result of compiling with InnoSetup then, to be copied from the InnoSetup Output directory to the upload directory.

Tarballs

Tarballs are to be prepared after compiling setup.exe, as we do not want to include Windows binaries.

On the Linux PC do

```
rm ~/tmp/viteSS3.4/MODULES/*
```

```
cp -p ~/ViteSSModulesUnix/* ~/tmp/viteSS3.4/MODULES/
```

Some library files help users if they want to compile own modules:

```
cp -p ~/SRC/vitess/trunk/SRC/g2-0.72/Linux_x86_64/libg2.a \  
~/tmp/vitess3.4/SRC/g2-0.72/Linux_x86_64/  
cp -p ~/SRC/vitess/trunk/SRC/g2-0.72/Darwin_x86_64/libg2.a \  
~/tmp/vitess3.4/SRC/g2-0.72/Darwin_x86_64/  
cd ~/tmp
```

The compression script pop produces tarballs or 7zballs:

```
~/SRC/vitess/trunk/TOOLS/pop vitess3.4  
~/SRC/vitess/trunk/TOOLS/pop vitess3.4 7
```

Putting distribution files in the target directory

Under Linux cd to an arbitrary directory where upload files should be., then

```
mv ~/tmp/vitess3.4.tar.gz ./  
mv ~/tmp/vitess3.4.7z ./
```

The script mkchecksums-txt.pl helps to prepare the checksums description file of the distribution.

```
~/SRC/vitess/trunk/TOOLS/mkchecksums-txt.pl 3.4 > checksums.txt.34
```

Upload Files

The result of all this are files to be uploaded to a web site / owncloud / nubes directory.

vitess3.4.tar.gz	normal tarball
vitess3.4.7z	7z compressed, small
vitess3.4.exe	Windows installer
checksums.txt.34	Text file to reflect checksums